Section 5.3: Positronium

A Study Guide and Rewrite of Griffiths Chapter 5.3

Based on *Introduction to Elementary Particles* by David Griffiths

1 What Is Positronium?

Hydrogen is an electron bound to a proton by the Coulomb force. But what if we replace the proton with *another* lepton? An **exotic atom** is any hydrogen-like bound state where one of the particles has been swapped out. Examples include:

System	Composition	Notes
Muonic hydrogen	$p^+\mu^-$	muon replaces electron
Pionic hydrogen	$p^+\pi^-$	pion replaces electron
Muonium	μ^+e^-	antimuon replaces proton
Positronium	$\boldsymbol{e^+e^-}$	**positron replaces proton**

Why positronium matters

Positronium (e^+e^-) is a purely leptonic atom—no quarks, no strong force, just QED. This makes it one of the cleanest systems for *precision tests of quantum electrodynamics*. It is also the direct model for **quarkonium** (Section 5.4), where the same logic is applied to quark–antiquark bound states.

Positronium was predicted theoretically by Pirenne in 1944 and first produced in the lab by Deutsch in 1951. It is unstable (the e^+ and e^- eventually annihilate), but it lives long enough to have a well-defined spectrum.

2 Energy Levels: The Reduced Mass Trick

2.1 The Problem: Two Equal Masses

In hydrogen, the proton is $\sim$1836 times heavier than the electron. We treat it as stationary and solve for the electron's motion. In positronium, both particles have the *same mass* m_e, so neither one is stationary—they both orbit their common center of mass.

The standard trick from classical mechanics: convert a two-body problem into an equivalent one-body problem using the **reduced mass**:

$$\boxed{\frac{1}{m_{\text{red}}} = \frac{1}{m_1} + \frac{1}{m_2}} \tag{1}$$

For hydrogen ($m_1 = m_e$, $m_2 = m_p \gg m_e$): $m_{\text{red}} \approx m_e$ (almost no change).
For positronium ($m_1 = m_2 = m_e$):

$$\boxed{m_{\text{red}} = \frac{m_e}{2}} \tag{2}$$

2.2 Consequences for the Bohr Formula

The unperturbed (Bohr) energy levels of hydrogen are:

$$E_n = -\frac{m_e}{2\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0}\right)^2 \frac{1}{n^2} = -\frac{13.6\text{ eV}}{n^2}$$

To get positronium, substitute $m_e \to m_e/2$:

$$\boxed{E_n^{(\text{Ps})} = -\frac{6.8\text{ eV}}{n^2}} \tag{5.27}$$

Key Result #1: Energy levels are halved

Every energy level of positronium is exactly half the corresponding hydrogen level. The ground-state binding energy is 6.8 eV instead of 13.6 eV.

2.3 Consequences for the Bohr Radius

The Bohr radius $a = \hbar^2/(me^2/4\pi\epsilon_0)$ scales as $1/m$. Since m_{red} is halved, the radius is *doubled*:

$$\boxed{a_{\text{Ps}} = 2a_0 = 1.06 \times 10^{-8}\text{ cm}} \tag{5.28}$$

Positronium is a bigger, more loosely bound atom than hydrogen.

3 Fine Structure of Positronium

Recall the hierarchy of corrections to hydrogen:

$$\underbrace{E_{\text{Bohr}}}_{\sim\alpha^2 mc^2} \gg \underbrace{E_{\text{fine}}}_{\sim\alpha^4 mc^2} \gg \underbrace{E_{\text{Lamb}}}_{\sim\alpha^5 mc^2} \gg \underbrace{E_{\text{hyperfine}}}_{\sim(m_e/m_p)\alpha^4 mc^2}$$

In hydrogen, hyperfine splitting is tiny because of the mass ratio $m_e/m_p \approx 1/1836$. But in positronium, this ratio is **1**. This has a dramatic consequence:

Key Result #2: Hyperfine ∼ Fine Structure

In positronium, the spin-spin interaction ("hyperfine" in hydrogen language) is the *same order* as the spin-orbit and relativistic corrections: all are $\sim \alpha^4 mc^2$. There is no separation of scales.

The complete fine-structure formula for positronium is:

$$E_{\text{f.s.}}^{(\text{Ps})} = \frac{\alpha^4 mc^2}{2n^3}\left[\frac{11}{32n} - \frac{\varepsilon}{(2\ell+1)}\right] \tag{5.29}$$

where $\varepsilon = 0$ for the **singlet** ($S = 0$) spin configuration, and for the **triplet** ($S = 1$):

$$\varepsilon = \begin{cases} -\dfrac{3\ell+4}{(\ell+1)(2\ell+3)}, & j = \ell+1 \\ \dfrac{1}{\ell(\ell+1)}, & j = \ell \\ \dfrac{3\ell-1}{\ell(2\ell-1)}, & j = \ell-1 \end{cases} \tag{5.30}$$

The Lamb shift ($\sim \alpha^5 mc^2$) makes only a small additional correction and is less interesting in positronium because the "accidental" degeneracy in ℓ is *already broken* at the fine-structure level.

4 Pair Annihilation: A New Feature

What's new in positronium that isn't in hydrogen?

The electron and positron can *temporarily annihilate* into a virtual photon, then re-materialize. This is a quantum fluctuation with no analog in hydrogen (the proton and electron can't annihilate each other).

4.1 The Feynman Diagram

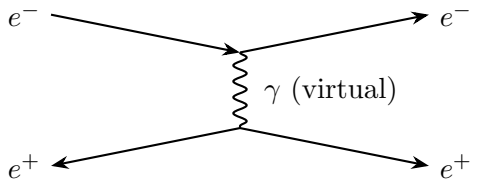


Figure 5.5: Virtual pair annihilation in positronium

4.2 When Does This Happen?

This diagram requires the e^+ and e^- to be *at the same point*. Two conditions:

1. **Only $\ell = 0$ states:** The wavefunction goes as r^ℓ near the origin, so $|\psi(0)|^2 \neq 0$ only when $\ell = 0$.

2. **Only triplet ($S = 1$):** The virtual photon has spin 1, so angular momentum conservation requires the triplet configuration.

The energy shift from this process is:

$$\boxed{\Delta E_{\text{ann}} = \frac{\alpha^4 mc^2}{4n^3}} \quad (\ell = 0,\ S = 1) \tag{5.31}$$

This is the *same order* as the fine structure ($\alpha^4 mc^2$), so it represents a significant correction.

5 The Energy Level Diagram

Figure 5.6 in Griffiths shows the complete splitting of the $n = 1$ and $n = 2$ Bohr levels after all corrections. Here is a schematic:

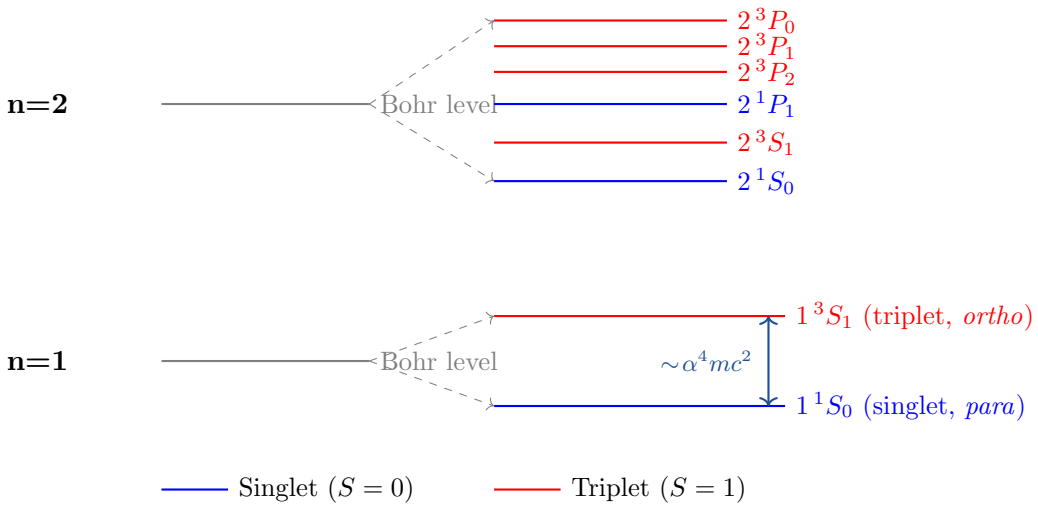


Figure 5.6 (schematic): Energy levels of positronium for $n=1$ and $n=2$

The notation $n\,^{2S+1}L_J$ is standard spectroscopic notation:

- n = principal quantum number
- $2S+1$ = spin multiplicity (1 for singlet, 3 for triplet)
- L = orbital angular momentum letter ($S, P, D, \ldots$ for $\ell = 0, 1, 2, \ldots$)
- J = total angular momentum ($\vec{J} = \vec{L} + \vec{S}$)

The singlet $1\,^1S_0$ state is called **para-positronium**. The triplet $1\,^3S_1$ state is called **ortho-positronium**.

6 Decay of Positronium

Unlike hydrogen, positronium is unstable—the e^+ and e^- eventually annihilate into real photons. The rules governing this decay are elegant.

6.1 The Selection Rule from Charge Conjugation

Positronium has a well-defined charge conjugation quantum number:

$$C_{\text{Ps}} = (-1)^{\ell+s}$$

A system of n photons has $C = (-1)^n$. For decay to be allowed, C must be conserved:

$$\boxed{(-1)^{\ell+s} = (-1)^n} \tag{5.32}$$

Since annihilation requires e^+ and e^- to overlap, it only happens from $\ell = 0$ states. This gives:

State	ℓ	s	$C = (-1)^{\ell+s}$	**Decays to**
Para-positronium ($^1S_0$)	0	0	$+1$	2γ (even # of photons)
Ortho-positronium ($^3S_1$)	0	1	-1	3γ (odd # of photons)

Key Result #3: Para decays fast, Ortho decays slow

Para-positronium decays to 2 photons and lives for about:

$$\tau_{\text{para}} \approx 1.25 \times 10^{-10} \text{ s}$$

Ortho-positronium decays to 3 photons. The extra factor of α (from the extra vertex in the Feynman diagram) makes it live about 1000 times longer:

$$\tau_{\text{ortho}} \approx 1.4 \times 10^{-7} \text{ s}$$

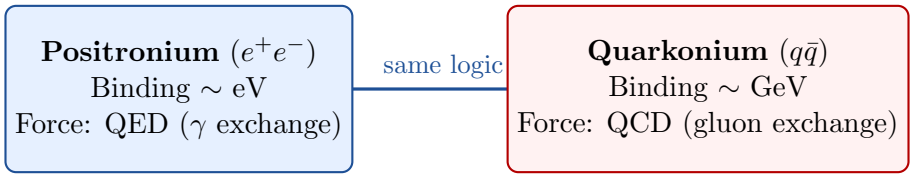
6.2 Why Not Decay to 1 Photon?

A single photon cannot conserve both energy *and* momentum simultaneously. In the center-of-mass frame, the initial momentum is zero, but a single photon always carries nonzero momentum $p = E/c$. So the minimum is 2 photons.

7 The Bridge to Quarkonium (Looking Ahead)

Why Griffiths puts positronium in this chapter

Section 5.3 is not just an exercise in atomic physics. It is the **template** for Section 5.4 on quarkonium. Every concept developed here—reduced mass, spin splittings, annihilation diagrams, spectroscopic labeling—carries over directly to the study of $c\bar{c}$ (charmonium) and $b\bar{b}$ (bottomonium) bound states. The energy level diagram of charmonium is *strikingly similar* to positronium, just scaled up by a factor of $\sim 10^8$ in energy.



Summary of Key Takeaways

1. **Positronium = hydrogen with equal masses.** Replace $m_e \to m_e/2$ (reduced mass). Energies halve; Bohr radius doubles.

2. **No scale separation.** Unlike hydrogen, the spin-spin coupling is the same order ($\alpha^4 mc^2$) as spin-orbit and relativistic corrections. Everything lumps together into "fine structure."

3. **Virtual annihilation is new.** The e^+e^- pair can briefly annihilate into a virtual photon—a process with no hydrogen analog. This shifts the triplet $\ell = 0$ states by an amount comparable to fine structure.

4. **Real annihilation obeys C-conservation.** Singlet ($C = +1$) $\to$ 2 photons (fast). Triplet ($C = -1$) $\to$ 3 photons (slow, suppressed by α).

5. **Positronium is the prototype for quarkonium.** The spectroscopic structure of charmonium and bottomonium mirrors positronium almost exactly, just at 10^8 times the energy scale. Understanding positronium is the gateway to understanding heavy mesons.